

Nathan Liu

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science & Data Science

May 2026

Berkeley, CA

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL), C/C++, TypeScript, Go

Backend/Data: Django, NumPy, Pandas, Matplotlib

Cloud/Infra: Kubernetes, AWS (Kinesis, Lambda, S3, EC2), Docker, GCP (BigQuery)

Machine Learning: PyTorch, Hugging Face, TensorFlow, LangChain

EXPERIENCE

Data Engineer Intern

May 2025 – Aug. 2025

Carbon Sustain

Berkeley, CA

- Assembled a real-time LLM decisioning pipeline on AWS to process 100k events/day which enforced schema-validated JSON outputs and wrote full traces to S3 for auditability and replay.
- Improved platform reliability by adding idempotency keys in DynamoDB and CloudWatch alarms to reduce failed executions and eliminated release-related deployment failures by 50%.
- Cut dashboard data retrieval time by 80% by rewriting high-cost joins, adding composite indexes, and implementing cursor pagination in Django/PostgreSQL APIs.

Software Engineer

Jan. 2025 – Aug. 2025

UnifIBD

Remote (San Jose, CA)

- Architected a mobile CI/CD pipeline with GitHub Actions, automating deployments to TestFlight + Google Play, which cut manual release times from hours to minutes.
- Streamlined cross-team development workflows between the analytics and engineering teams, reducing feature handoff latency by half and accelerating time-to-production for data-driven updates.
- Built a Django backend and admin CMS that enabled marketing to update in-app content without requiring an app store submission.

Data Science Intern

Jun. 2024 – Aug. 2024

L.A. Lucky Import & Export Inc.

Los Angeles, CA

- Reconstructed a near-real-time SAP inventory sync for 3,647+ SKUs using incremental pulls and upserts into Postgres which reduced manual entry by 100% and enabled automated invoicing for all customers.
- Overhauled a legacy platform (Django REST, JWT, React) and improved performance by 75% by reducing payload size and adding server-side caching.
- Built a Docker-based CI/CD workflow to deploy ETL/ELT jobs automatically, streamlining legacy database migration and improving pipeline reliability.

PROJECTS

Live Semantic Market | *JavaScript*

- Engineered a streaming data pipeline processing and transforming global research feeds via SSE, projecting multi-dimensional vector drifts with sub-100ms latency.
- Built an embedding pipeline that vectorizes unstructured academic papers and computes cosine similarity matrices to price abstract AI concepts in real time.

Clickbait Classifier | *Python, PyTorch, Pandas*

- Fine-tuned a BERT-based classifier on 18K social-news titles; achieved 0.89 F1 and 91% accuracy on a held-out test set, outperforming a TF-IDF logistic regression baseline by +0.14 F1 points.

ItineraryAI | *Python, LangChain, FastAPI, JavaScript*

- Built a chat-based travel concierge that generates structured itineraries from a single prompt and shipped a web dashboard for trip planning (budgets, maps, calendar sync).